

Spine Metrix

User Tutorial

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. Experimental / educational and research tool — not a medical device.

1. Introduction

The **Spine Metrix** is a platform for sagittal and coronal spine analysis and surgical planning from radiographs/DICOM, with a Magnetic Resonance Imaging (MRI) module. This tutorial shows, step by step, how to operate the program: from opening an image and marking vertebrae to simulating the surgical correction and exporting the report.

The program interface is entirely in **English**; therefore, throughout the tutorial, the button and menu names appear in English (e.g., **Open File**, **Mark**, **Full Analysis**), with the explanation in English.

Before you start

This is a document to support the use of the software. Spine Metrix is an experimental, educational and research tool — it is not a medical device. Every measurement or simulation is an estimate and must be confirmed by standard clinical methods and by the judgment of a qualified professional. For the clinical meaning of the parameters (PI, PT, SVA, GAP, etc.), see the Clinical Manual, the companion document to this tutorial.

2. The Interface

The Spine Metrix screen has four main areas. It is worth recognizing them before starting the work.



Area	What it is
Top menu bar	File / Mode / Mark / Measure / Analysis / View — the same actions as the side bar. The active mode is marked with a dot (●) in the menu.
Side bar	The tools of the mode you are currently working in.
Central area	The image (or the two images, in Biplanar), on a black background.
Minimap / thumbnail	Navigation thumbnail; in Biplanar, a schematic diagram of the cages and the rod.

Tip — Hide Image

The **Hide Image** button (radiograph icon) hides the X-ray and shows only the drawings and the bars over the black background. It is very useful for reading the charts and analysis lines without the image underneath.

3. Choosing the Working Mode (Mode)

The **Mode** menu defines which workflow you are in. Each mode has its own purpose and tools.

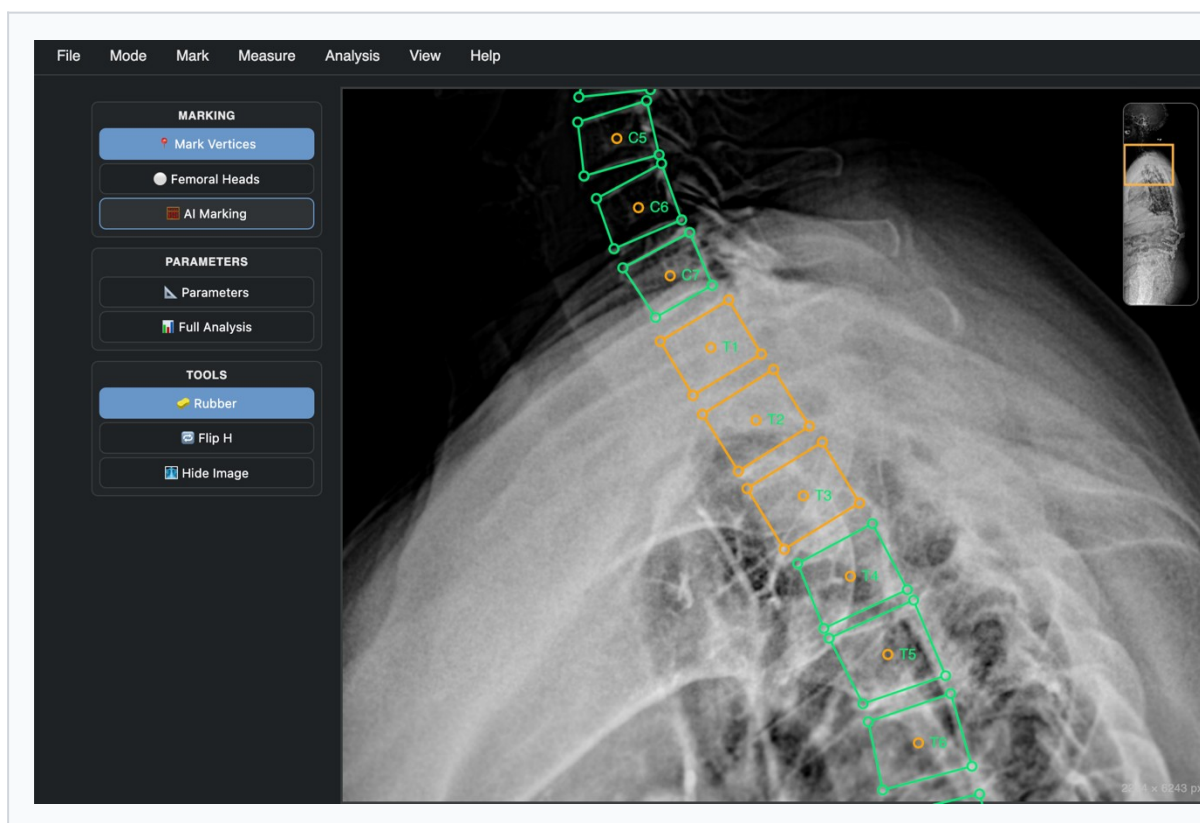
Mode	For what
Sagittal	Profile (lateral): mark vertebrae, count levels and obtain the sagittal parameters (PI, PT, SS, SVA, LL, TK).
Coronal	Frontal (AP): mark vertebrae and analyze scoliosis (Cobb by inflections).
Photogrammetry	Photo (non-radiographic) with perspective correction.
Biplanar	Surgical planning on the sagittal and coronal views simultaneously — the heart of the program.
MRI	Magnetic Resonance Imaging viewer: two series, cross-reference lines, measurements and VBQ.

The **Open File** and **Save** commands are in the **File** menu (and also in the side bar). When opening an image, choose the **plane** (Sagittal or Coronal) — each plane keeps its own markings.

4. Sagittal Workflow (Profile)

This is the basic workflow to obtain the sagittal parameters of a profile radiograph. Follow the steps in order.

- 1. Open File:** select the sagittal X-ray or DICOM (Sagittal plane).
- 2. Metric calibration** (optional, but required for measurements in mm): use the metric reference of the image. The DICOM reference (PixelSpacing) is read automatically.
- 3. Mark → Femoral Heads:** mark the two femoral heads. They define the hip axis and the anterior/posterior orientation. Without them, PI, PT and SS — and the image orientation itself — may be undefined.
- 4. Mark vertebrae:** mark the four corners of each visible vertebra, from S1 to C2. You do not need to mark them all — only the clear ones. In practice, you can mark from S1 up to the region near T1, stop where visibility decreases, and resume from C2 going down to that same less-visible region. The program has a Marking AI that can complete the marking of up to 6 vertebrae automatically (the less-visible thoracic region).
- 5. Marking AI** (Marking group): predicts the missing vertebrae, shown in orange. Touching a corner of an orange vertebra promotes it to green (user marking). It is an action, not a toggle: each click re-runs the analysis.
- 6. Parameters** (toggle): turns on the centroid spline, the sagittal parameters and the chart.
- 7. Analysis → Full Analysis:** opens the "Degree of Curvature" chart (curvature by level).
- 8. Patient Age** (Analysis): enter the age to enable the age-adjusted targets (Lafage).
- 9. Import .pkl** (optional, shortcut Shift+P): imports legacy markings from the desktop.



5. Coronal Workflow (AP)

1. **Mode** → **Coronal** and **Open File**: open the coronal X-ray (Coronal plane).
2. Mark the vertebrae (four corners) with the label of each level. The label helps match the coronal with the sagittal in Biplanar. It is not necessary to mark every vertebra, only those that span the curvatures you want to analyze or those needed for the 3D analysis (usually from C4 to S1).
3. The coronal analysis (Cobb by inflection points) is in the **Analysis** menu.

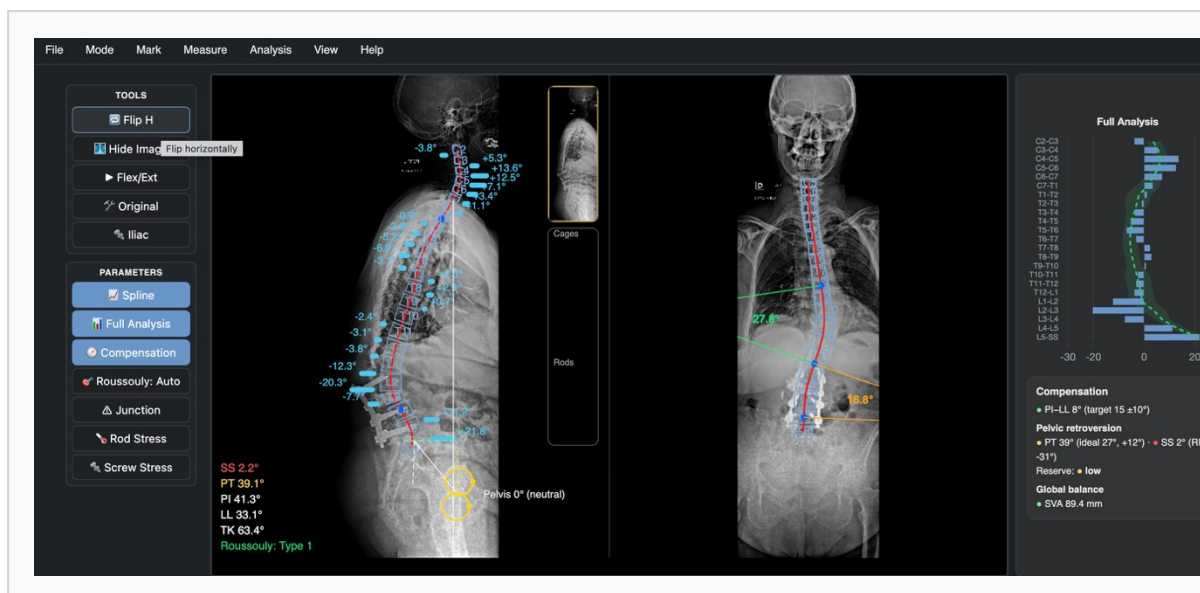
Tip — matching coronal with sagittal

The program matches the two planes **by the label** of each level. If some level does not match in Biplanar, check that the labels are correct in both planes (in the absence of a label, matching is done by proportional height).

6. Biplanar Workflow — the 3 Moments

The **Biplanar** mode is where surgical planning happens, on the sagittal and coronal views at the same time. **Prerequisites:** have the Sagittal and the Coronal already marked, and the femoral heads marked on the sagittal.

When you enter **Mode** → **Biplanar**, the two images appear (sagittal on the left, coronal on the right). The program starts at the **Original** moment. The mode button cycles through the three moments: Original → Planning → Simulation → Original.



6.1 Original Moment (read)

- Shows only the two X-rays, without drawings.
- Full Analysis:** vertebrae in blue on both sides, plus the chart and the bars. It hides the X-ray — which can be brought back with **Show Image**, keeping the vertebrae on top.
- Compensation:** adds the pelvic references (PT/PI), the SVA, the femur and the analysis box on the right.
- The text on the right panel is single — it always shows that of the last button pressed.

6.2 Planning Moment (correct)

When you enter Planning, the program already turns on Compensation and Spline, keeps **Show Image** and shows only the screws (without rods or connections). From here you instrument and correct.

Instrument and select

- Instrument (double click):** on a vertebra inserts a screw; on a disc inserts a cage; on the tulip removes the screw. In the coronal view, screws appear bilateral (two per vertebra, tilted as in the AP).
- Select:** click a vertebra (sagittal) to select it. The ↑/↓ arrows navigate between levels; the ←/→ arrows rotate the selected vertebra (chain osteotomy — the block above follows).
- Pelvis:** click a femoral head, or navigate with ↑/↓ to S1, to select the pelvis. The ←/→ arrows rotate the pelvis (with PI constant).

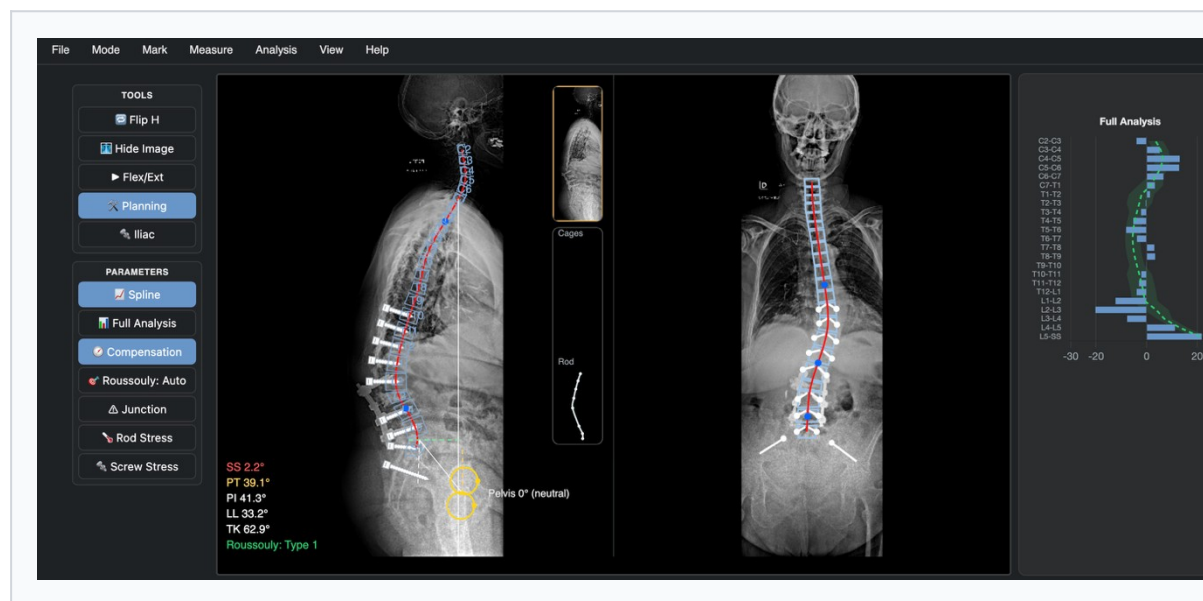
Correct

- Sagittal correction (corS):** Shift + ←/→ reduces the spine to the target rod (the desired lordosis, distributed according to Roussouly). The **Roussouly** button cycles the target type (Auto / 1 / 2 / 3 / 4).

- **Coronal correction (corC):** click the coronal image (selects all vertebrae) and use ←/→ to straighten the scoliosis; the coronal X-ray slides to follow. The **0** key resets.
- **Screw color:** white with Show Image; lilac with Hide Image.

Analysis buttons (toggles)

Available in Planning: **Full Analysis**, **Compensation**, **Junction** (junctional angles PJA/DJA), **Rod Stress** (rod stress), **Screw Stress**, **Spline**, **Flex/Ext** (flexion-extension animation) and **Iliac** (iliac screw, by two clicks).



6.3 Simulation Moment (animate)

- Shows the 3D on the coronal side. The pelvis appears already corrected, the spine and screws in the original, and the rods in the corrected position (the lilac target bar).
- The ←/→ arrows animate the spine from the original to the rod — sagittal, 3D and coronal together.
- It is visualization only: it does not change the plan.

Tip — what the Simulation is for

The Simulation is the way to visually check the result of the plan before exporting: you see the transition from the original spine to the corrected one. Since it changes nothing, use it freely to demonstrate or review.

7. Export and Save

- **Save Annotation / Save Image** (File menu): saves the markings (on the computer and on the server, for research) and the image.
- **Save Report (PDF)**: generates the report with the parameters and the chart.
- **Persistence**: the markings are recovered automatically when the case is reopened. There are safety nets to prevent data loss — if the local storage is empty, the program fetches from the server.

8. Keyboard Shortcuts (main)

Key	Action
z / p / w	Zoom / Pan / Window-Level (brightness and contrast)
r	Reset the view
s	Sacral mode (sacrum marking)
H	In the .pkl preview, toggles the horizontal flip
Shift + P	Import .pkl markings (Sagittal)
← / →	Rotate the selected vertebra or pelvis · (coronal) adjust the coronal correction · (Simulation) animate
↑ / ↓	Change the selected vertebra (on reaching S1, selects the pelvis)
Shift + ← / →	Sagittal correction (reduction to the rod) · Shift+0 resets
Double click	Instrument: vertebra = screw, disc = cage, tulip = remove

Confirm the keys in the published version of the program; some tools are also available through the menus.

9. Tips and Troubleshooting

Situation	What to do
I want to read the bars and the chart clearly	Use Hide Image to see the bars over the black background.
The image is flipped (anterior on the right)	Mark the femoral heads — the program infers the orientation and aligns the chart with the Roussouly line. Without the femur, it assumes the default orientation.
PI, PT and SS do not appear	The femoral heads are missing. Mark them (Mark → Femoral Heads).
The age-based targets are blank	Enter the age (Analysis → Patient Age) and calibrate the millimeters.
The coronal does not match the sagittal	Check the labels of each level — matching is by label; failing that, by proportional height.
I deleted markings by mistake	For safety, the program does not overwrite existing markings with an empty state. Reload the case to bring back the previous markings.

10. MRI Module (summary)

- **MRI Viewer:** two series side by side, with cross-reference lines and measurements (Length / Slope / Angle).
- **MRI Findings** (Analysis menu): structured documentation of the findings by lumbar disc.
- **Assisted VBQ** (Vertebral Bone Quality): bone-quality estimate from the MRI signal.

The clinical detail and the formulas of the MRI module (including the VBQ) are in the Clinical Manual, the companion document to this tutorial.

Spine Metrix user tutorial — a document to support the use of the software, subject to updates as new versions are released. Authors: Arlindo Neto Montagnoli and Adimilson dos Santos Delgado · Supervision: Prof. Helton Luiz Aparecido Defino · Collaboration: Claude (Anthropic, 2026).